

University of Amsterdam at the CLEF 2026 SimpleText Track

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Jan Bakker and Jaap Kamps

Task 1: Simplify Scientific Text

Two main datasets

- 1) Cochrane-auto (Bakker and Kamps., 2024)
- 2) Cochrane-sections (Bakker and Kamps, 2025)

Technical abstract

Background

Surgeons and their assistants are especially at risk of exposure to blood due to glove perforations and needle stick injuries during operations. The use of blunt needles can reduce this risk because they don't penetrate skin easily but still perform sufficiently in other tissues.

Objectives

To determine the effectiveness of blunt needles compared to sharp needles for preventing percutaneous exposure incidents among surgical staff.

Search methods

We searched MEDLINE and EMBASE (until May 2011), CENTRAL, NHSEED, Science Citation Index Expanded, CINAHL, Nioshtic, CISdoc, PsycINFO, and LILACS (until September 2010).

Selection criteria

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Data collection and analysis

Two authors independently assessed study eligibility and risk of bias in trials and extracted data. We synthesized study results with a fixed-effect model meta-analysis.

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We located 10 RCTs involving 2961 participating surgeons performing an operation in which the use of blunt needles was compared to the use of sharp needles. Four studies focused on abdominal closure, two on caesarean section, two on vaginal repair and two on hip replacement. On average, a surgeon that used sharp needles sustained one glove perforation in three operations. The use of blunt needles reduced the risk of glove perforations with a relative risk (RR) of 0.46 (95% confidence interval (CI) 0.38 to 0.54) compared to sharp needles. The use of blunt needles will thus prevent one glove perforation in every six operations.

In four studies, the use of blunt needles reduced the number of self-reported needle stick injuries with a RR of 0.31 (95% CI 0.14 to 0.68). Because the force needed for the blunt needles is higher, their use was rated as more difficult but still acceptable in five out of six studies.

The quality of the evidence was rated as high.

Authors' conclusions

There is high quality evidence that the use of blunt needles appreciably reduces the risk of exposure to blood and bodily fluids for surgeons and their assistants over a range of operations. It is unlikely that future research will change this conclusion.

Plain language summary

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Two main approaches

- 1) Plan-Guided Text Simplification (Cripwell et al., 2023)
- 2) Retrieval-Augmented Text Simplification (Guo et al., 2023; Mathas et al., 2026)

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Plan-guided simplification model:
Qwen2.5-32B-Instruct

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Given an abstract section that requires simplification, retrieve similar sections.

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Best-of- N method

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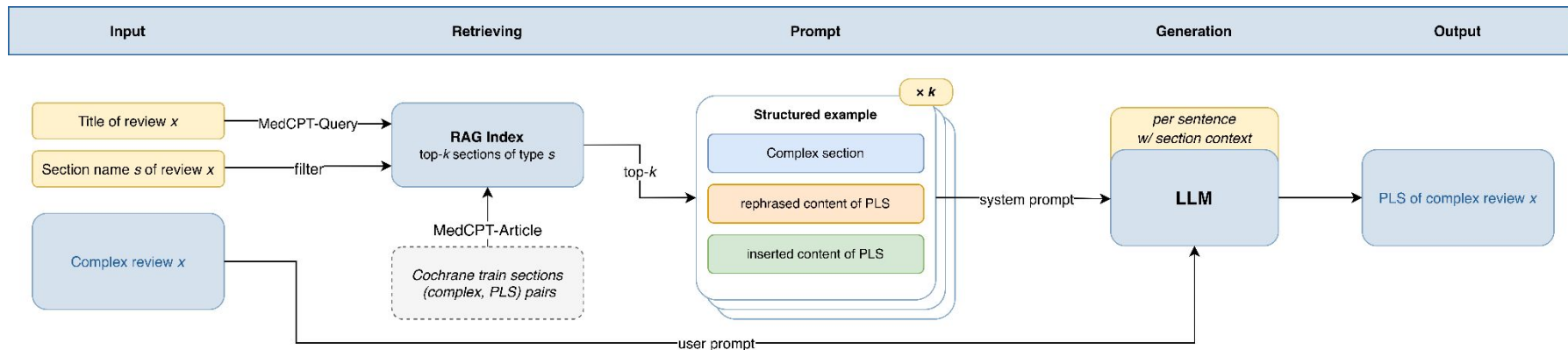
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There is high quality evidence that the use of blunt needles appreciably reduces the risk of exposure to blood and bodily fluids for surgeons and their assistants over a range of operations. It is unlikely that future research will change this conclusion.

Plain language summary

Surgeons and their assistants are especially at risk of needle stick injuries during operations. This can lead to infection with HIV or other blood-borne viruses. The use of blunt needles is proposed to prevent needle stick injuries. We reviewed the literature to evaluate the preventive effect of blunt needles compared to sharp needles on needle stick injuries among surgical staff. We searched multiple medical databases (to May 2011). We included studies if they were randomised controlled trials (RCTs) of blunt versus sharp suture needles for preventing needle stick injuries among surgical staff. We located 10 RCTs with 2961 operations in which blunt needles were compared to sharp needles. Six studies focused on abdominal operations, two on vaginal repair and two on hip replacement. On average, a surgeon that used sharp needles sustained one glove perforation per three operations. The use of blunt needles reduced the risk of glove perforations by 54% (95% confidence interval 46% to 62%) compared to sharp needles. The use of blunt needles in six operations will thus prevent one glove perforation. In four studies the use of blunt needles also reduced the number of self-reported needle stick injuries by 69% (95% confidence interval 14% to 68%). Even though surgeons reported that the force needed for the blunt needles was higher, their use of the needles was still rated as acceptable in five out of six studies. We concluded that there is high quality evidence that the use of blunt needles appreciably reduces the risk of contracting infectious diseases for surgeons and their assistants over a range of operations by reducing the number of needle stick injuries. It is unlikely that future research will change this conclusion.

Retrieval-Augmented Simplification



Task 1: Simplify Scientific Text

Evaluation

Run

RAG_BoN_k4_Q32L70

RAG_BoN_k4_Q32

PG_PCW_Q32

LLM_Q32

RAG_k4_Q32

Task 1: Simplify Scientific Text

Evaluation against simple_original

Run	BLEU $\uparrow$	SARI $\uparrow$	FKGL $\downarrow$
RAG_BoN_k4_Q32L70	12.80	47.57	12.34
RAG_BoN_k4_Q32	11.07	46.67	11.00
PG_PCW_Q32	12.51	46.50	10.37
LLM_Q32	9.41	46.20	10.55
RAG_k4_Q32	9.90	46.11	10.81

Task 1: Simplify Scientific Text

Evaluation against simple_full

Run	Back-ground	Objectives	Search methods	Selection criteria	Data coll. & analysis	Main results	Authors' conclusions	Mean
RAG_BoN_k4_Q32L70	44.83	40.15	37.59	40.16	34.53	44.22	44.99	40.93
RAG_BoN_k4_Q32	44.07	40.93	33.85	39.09	33.77	44.00	43.94	39.95
RAG_k4_Q32	43.96	40.64	33.53	39.08	33.71	43.79	43.63	39.76
LLM_Q32	43.46	37.36	5.17	23.80	32.85	43.78	43.82	32.89
PG_PCW_Q32	43.99	36.62	7.32	23.57	12.77	43.92	43.26	30.21

Task 1: Simplify Scientific Text

System evaluation using LENS and LLM-as judge, on CLEF 2026 (only English), sorted by LENS.

Run	LENS μ	LENS σ	Llama-3.1-8B. μ	Phi-3-mini-4k. μ	Qwen2.5-7B. μ
RAG_BoN_k4_Q32L70	68.31	5.61	67.64	38.52	72.27
LLM_Q32	67.50	8.11	60.23	51.80	73.62
RAG_k4	65.71	7.45	62.97	46.04	72.61
RAG_BoN_k4_Q32	64.99	7.76	66.11	44.12	72.39
PG_PCW_Q32	57.39	9.20	49.31	43.90	69.33

Task 2: Controlled Creativity

Run	Source	Model	Description
UvA_Task22a_GPT41Mini_nosource.zip	No	GPT-4.1-mini	Zero-shot no-source prompt
UvA_Task22a_GPT41Mini_PrecisionV2.zip	No	GPT-4.1-mini	Precision-oriented no-source prompt
UvA_Task22a_Qwen3_14B_nosource.zip	No	Qwen3-14B	Zero-shot open-weight no-source prompt
UvA_Task22b_GPT41Mini_GroundedV1.zip	Yes	GPT-4.1-mini	Zero-shot source-aware prompt

Task 2: Controlled Creativity

Run	Source	Model	Description
UvA_Task22a_GPT41Mini_nosource.zip	No	GPT-4.1-mini	Zero-shot no-source prompt
UvA_Task22a_GPT41Mini_PrecisionV2.zip	No	GPT-4.1-mini	Precision-oriented no-source prompt
UvA_Task22a_Qwen3_14B_nosource.zip	No	Qwen3-14B	Zero-shot open-weight no-source prompt
UvA_Task22b_GPT41Mini_GroundedV1.zip	Yes	GPT-4.1-mini	Zero-shot source-aware prompt

Run	Doc. F1	Prec.	Rec.	Sent. F1	Cat. Acc.
UvA_Task22a_GPT41Mini_nosource.zip	0.4291	0.3220	0.6428	0.3980	0.4178
UvA_Task22a_GPT41Mini_PrecisionV2.zip	0.4339	0.3236	0.6584	0.4113	0.4454
UvA_Task22a_Qwen3_14B_nosource.zip	0.4056	0.2718	0.7989	0.3351	0.3463
UvA_Task22b_GPT41Mini_GroundedV1.zip	0.4061	0.2758	0.7695	0.3117	0.5032

Task 3: Research Area Classification

Task Run	Description
3	UAmsterdam_1 Majority class baseline (Natural Sciences, Physics).
3	UAmsterdam_2 Zero-shot LLM, Gemini 3.5 Flash (each test title and abstract individually, with the full taxonomy).

Task 3: Research Area Classification

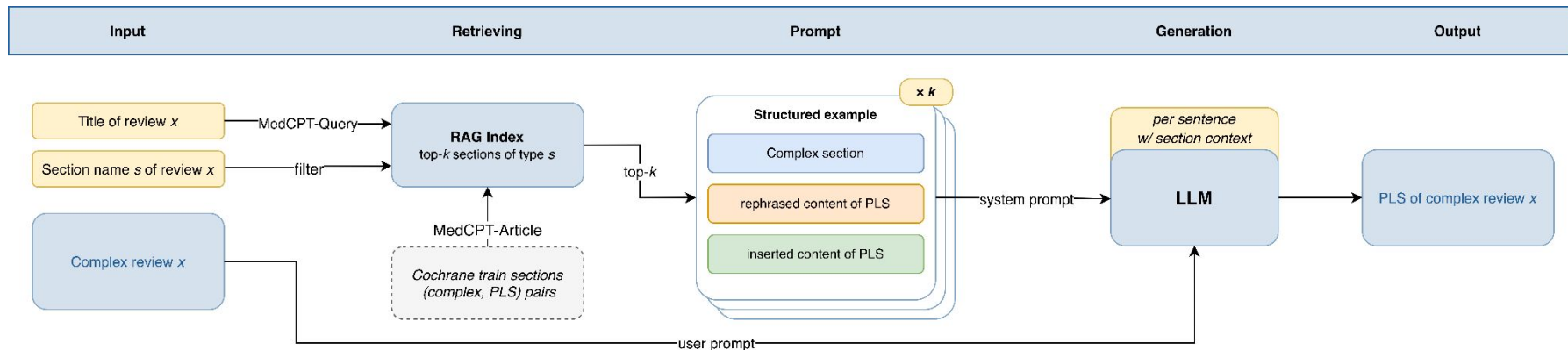
Task Run	Description
3	UAmsterdam_1 Majority class baseline (Natural Sciences, Physics).
3	UAmsterdam_2 Zero-shot LLM, Gemini 3.5 Flash (each test title and abstract individually, with the full taxonomy).

Team Name	Discipline Area			Field Area		
	EM	SD	ED	EM	SD	ED
UAmsterdam_2	0.6122	0.7035	0.7759	0.7033	0.8279	0.8472
UAmsterdam_1	0.0625	0.2281	0.4099	0.2000	0.6225	0.6659

Summary

Task 1: Baseline, plan-guided and retrieval-augmented simplification. **Task 2 & 3:** Zero-shot baselines.

Main conclusion: Retrieval-Augmented Simplification leads to the best section-level SARI scores.



Thank you for listening

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